

Initial Date: 07/19/2023
Revised Date: 12/10/2025

Section: 9-24R

Fentanyl

Pharmacological Category: Analgesic, Opioid; General Anesthetic

Routes: IV/IO/IM/IN

Indications:

1. Pain management
2. Patient sedation

Contraindications:

1. Altered Mental Status
2. Hypotension
3. Respiratory Depression

Expected effects:

1. Decreased pain
2. Decreased agitation

Side effects:

1. Drowsiness
2. Hypotension
3. Respiratory Depression
4. Vomiting

Dosing: CHEST PAIN/ACUTE CORONARY SYNDROME

Indication: Chest pain in which nitroglycerin is contraindicated due to erectile dysfunction medication or suspected cardiac chest pain is refractory to nitroglycerin.

Adults administer 1 mcg/kg IV/IO/IM/IN dosage rounded as per below:

- a. Up to and including 25 kg - administer 25 mcg
- b. 26-50 kg – administer 50 mcg
- c. 51-75 kg - administer 75 mcg
- d. 76 kg and above – administer 100 mcg
- e. IV/IO/IN may repeat one-time, total dose may not exceed 200 mcg
- f. If IM administration may NOT repeat

Dosing: PAIN MANAGEMENT

Indication: Patient is unable to tolerate ketamine or ketamine is not available and the patient has significant pain (described as 7 or greater on the Wong Pain Scale).

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Adults (patients > 14 years of age), administer 1 mcg/kg IV/IO/IM/IN dosage, rounded as per below:

- a. Up to and including 25 kg - administer 25 mcg
- b. 26-50 kg – administer 50 mcg
- c. 51-75 kg - administer 75 mcg
- d. 76 kg and above – administer 100 mcg
- e. IV/IO/IN may repeat one-time, total dose may not exceed 200 mcg. Interval between doses is minimally 10 minutes.
- f. If IM administration may NOT repeat

Pediatrics administer:

1. Fentanyl according to MI MEDIC
2. If MI MEDIC is not available administer:
 - a. Fentanyl 1 mcg/kg IV/IO/IM/IN
 - b. May repeat IV/IO/IN one time. Interval between doses is minimally 10 minutes.
 - c. If IM administration may NOT repeat

Dosing: PATIENT PROCEDURAL SEDATION

Adults administer:

1. Fentanyl 50-100 mcg (1 mcg/kg) IV/IO titrated slowly (IN, if available). May repeat every 4 minutes to a maximum of 3 mcg/kg.

Pediatrics administer:

1. Fentanyl according to MI MEDIC
2. If MI MEDIC is not available administer:
 - a. Fentanyl 1 mcg/kg IV/IO titrated slowly (IN, if available). May repeat every 5 minutes to a maximum of 3 mcg/kg.

Used in the Following Protocols

Chest Pain/Acute Coronary Syndrome (Section 5 Adult Cardiac)

Pain Management (Section 7 Procedures)

Patient Procedure Sedation (Section 7 Procedures)